configurationManagement

Service Description

Service ID: *"configurationManagement"*

Abstract
This document describes the `configurationManagement` service, version 1.0.0, produced by the Configuration support system of Arrowhead (documentation edition 5.3.0).

Contents

1 Documentation edition and implementation scope **3**

2 Overview **3**
2.1 Significant Prior Art . . . 3
2.2 How This Service Is Meant to Be Used . . . 3
2.3 Important Delimitations and Dependencies . . . 3

3 Service Interface **4**
3.1 operation list-configurations . . . 4
3.2 operation store-configuration . . . 4
3.3 operation remove-configuration . . . 4

4 Information Model **5**
4.1 struct ConfigurationListResponse . . . 5
4.2 struct ConfigurationRequest . . . 5
4.3 struct ConfigurationResponse . . . 5
4.4 Primitives . . . 5

5 References **7**

6 Revision History **8**
6.1 Amendments . . . 8
6.2 Quality Assurance . . . 8

1 Documentation edition and implementation scope

This is the 5.3.0 documentation edition, dated 2026-11-09. It carries forward the system and service contracts from the retained 5.2.2 documentation and identifies the current primary implementation sources. The former `documentation/v5.2.2` folder is retained unchanged.

Document version 5.3.0 is distinct from executable, dependency, registered system/service and wire-contract versions. Values still defined by the source (including 5.2.2 artefact versions, 1.0.0 service versions and `SysMonitor-AMQP-IDD-5.2.2` payload identifiers) are preserved. This documentation update does not change runtime behaviour or establish a new implementation release. Historical verification statements describe the tests recorded in the inherited baseline; this revision validates document compilation and references.

2 Overview

This document describes the `configurationManagement` service, which provides operator management of all stored configurations. It is produced by the Configuration system [1] in service version 1.0.0.

The rest of this document is organized as follows. In Section 3, we describe the abstract message operations provided by the service. In Section 4, we end the document by presenting the data types used by the mentioned operations. The concrete interface is specified in the HTTP/TLS/JSON IDD [2].

2.1 Significant Prior Art

Successor of the Arrowhead 4.6 Configuration management endpoints.

2.2 How This Service Is Meant to Be Used

Operators list, store and remove configurations.

2.3 Important Delimitations and Dependencies

Files are validated (Base64, media type, plain file name, `configuration.max.size`). Access follows the management policy (standalone deployment: `sysop-only`).

3 Service Interface

This section describes the interfaces to the `configurationManagement` service. In particular, each subsection names an abstract operation, an input type and an output type, in that order. The input type is named inside parentheses, while the output type is preceded by a colon. Input and output types are only denoted when accepted or returned, respectively, by the interface in question.

All abstract data types named in this section are defined in Section 4.

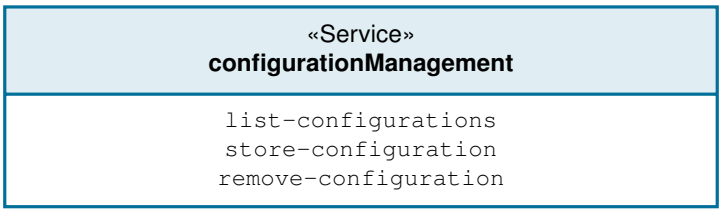


Figure 1: SysML-style block of the `configurationManagement` service and its operations.

The following interface operations are available.

3.1 operation list-configurations () : ConfigurationListResponse

Returns all configurations.

3.2 operation store-configuration (ConfigurationRequest) : ConfigurationResponse

Stores the configuration of the named system.

3.3 operation remove-configuration (Name) : ConfigurationResponse

Removes the configuration of the named system and returns it.

4 Information Model

Here, all data objects that can be part of the `configurationManagement` service are listed in alphabetic order. Note that each subsection, which describes one type of object, begins with the *struct* keyword, which is used to denote a collection of named fields, each with its own data type. As a complement to the explicitly defined types in this section, there is also a list of implicit primitive types in Section 4.4, which are used to represent things like hashes and identifiers.

4.1 struct ConfigurationListResponse

List of configurations.

Field	Type	Description
count	Number	Total number of matching records (all pages).
data	List<ConfigurationResponse>	The stored configurations.

4.2 struct ConfigurationRequest

A configuration file to store.

Field	Type	Description
systemName	Name	System the configuration belongs to (proxy: required; management: taken from the path).
fileName	String	Plain file name of the configuration.
contentType	String	Media type of the configuration file.
data	String	File content, Base64 (at most `configuration.max.size` bytes decoded).

4.3 struct ConfigurationResponse

A stored configuration file.

Field	Type	Description
id	Number	Identifier of the stored configuration.
systemName	Name	System the configuration belongs to.
fileName	String	Plain file name of the configuration.
contentType	String	Media type of the configuration file.
data	String	File content, Base64.
size	Number	Size of the decoded file in bytes.
createdAt	DateTime	UTC time of creation.
updatedAt	DateTime	UTC time of the last change.

4.4 Primitives

Types and structures mentioned throughout this document that are assumed to be available to implementations of this service. The concrete interpretations of each of these types and structures must be provided by any IDD document claiming to implement this service.

Type	Description
DateTime	Pinpoints a specific moment in time (UTC).
List	An *array* of a known number of items, each having type A.
Name	A string identifier that is intended to be both human and machine-readable.
Number	An integer number.
String	A chain of characters.

5 References

[1] J. Delsing, “Configuration – System Description,” Eclipse Arrowhead, SysD 5.3.0, 2026, ah5-Configuration-java-spring/documentation/v5.3.0/Configuration-v5.3.0-SysD.pdf.

[2] ——, “configurationManagement HTTP/TLS/JSON – Interface Design Description (Configuration),” Eclipse Arrowhead, IDD 5.3.0, 2026, configuration-v5.3.0-configurationManagement-HTTP-IDD.pdf.

6 Revision History

6.1 Amendments

No.	Date	Version	Subject of Amendments	Author
1	2026-10-22	5.2.2	Document created from the Arrowhead LaTeX templates for the 5.2.2 Java Spring implementation, including the local extensions of this workspace.	Jerker Delsing
2	2026-10-30	5.2.2	Checked against the primary Java source tree (Eclipse-Arrowhead-java): standalone deployment, configuration and security statements updated.	Claude Opus 5.5 (draft)
3	2026-11-09	5.3.0	New documentation edition; primary-source references and Go/Rust implementation scope; 5.2.2 retained.	Documentation maintenance

6.2 Quality Assurance

No.	Date	Version	Approved by
1		5.3.0	